GCSC WHITEPAPER v1.1

Global Construction Smart Contract

Decentralized ecosystem for the 3T construction industry

Hybrid Credit Infrastructure · AI Credit Engine · External Brain

GCSC DAO · April 2026 · gcscdao@gmail.com

Section 01

Abstract

The global construction industry generates approximately **3 trillion** in annual output yet operates on antiquated trust mechanisms. GCSC (Global Construction Smart Contract) is a DAO + DeFi protocol purpose-built for the construction vertical, deployed on **XPR Network (Proton)**. GCSC v1.1 introduces three core pillars: Hybrid Credit Infrastructure (regulated + DeFi lending at 0.5–2% APR), AI Credit Engine (automated ML-based underwriting with explainability), and External Brain (LightRAG + Knowledge Graph + MCP for persistent agent memory).

Section 02

Problem Statement

Pain Point	Industry Impact
Payment delays 30–90 days	~–2T annual loss from cash-flow gaps
Fraudulent contractors	1B/year homeowner losses (unlicensed)
Zero worker benefits	7.5M US construction workers, 0% retirement coverage
Lost institutional knowledge	Project history disappears when PMs leave

Section 03

Hybrid Credit Infrastructure

GCSC bridges regulated lending (licensed partners, KYC/AML compliant) with DeFi liquidity pools (GCST + USDC). Loans disburse in <60 seconds via mppx-xpr-network machine payments at 0.5–2% APR — vs. 18–29% on credit cards.

Tier	Max Loan	APR	Collateral
Tier 1 (Basic)	0,000	0.5–1.0%	Equipment NFTs
Tier 2 (Standard)	50,000	1.0–1.5%	Equipment NFTs + Staked GCSC
Tier 3 (Premium)	00,000	1.5–2.0%	Real Estate DAO + NFTs

Revenue split: 70% lenders · 20% Treasury · 10% GCSC buyback & burn

Section 04

AI Credit Engine

The Risk Assessment Agent (RAA) combines on-chain signals (payment history 35%, project completion 25%, equipment collateral 15%, license status 15%, staking 10%) with an XGBoost ML model for 12-month default prediction. Every decision includes SHAP-based XAI explanation for regulatory compliance and contractor appeals.

Section 05

External Brain — LightRAG & Knowledge Graph

LightRAG indexes all GCSC documents and decisions with <200ms query latency. A Neo4j knowledge graph stores entities and relationships across contractors, projects, permits, and disputes. MCP (Model Context Protocol) enables persistent agent memory across sessions — no decision is ever lost.

Section 06

Tokenomics — GCSC & GCST

Dual Token Model: GCSC (governance + utility, max 1B supply) + GCST (stablecoin pegged , backed 50% USDC + 30% staked GCSC + 20% real assets).

Allocation	%	Notes
Ecosystem Rewards	40%	10-year declining emission
Treasury / RE DAO	20%	Locked until RE DAO launch
Team & Advisors	15%	4yr vest, 1yr cliff
Initial Liquidity	10%	DEX bootstrap
Community Sale	10%	Public sale
Reserve Fund	5%	Emergency buffer

Section 07

Roadmap

Phase	Timeline	Key Milestones
Foundation	M1–M3	Token launch, core contracts, WebAuth, MVP app, External Brain MVP
Growth	M4–M8	CMA + RAA agents live, Proton Loan, 100 contractors, Equipment NFTs, DEX
Expansion	M9–M18	Real Estate DAO, 5 US states, Metal Bridge, CA agent, 500+ workers
Maturity	M19–M36	Full AI autonomy, 10K+ contractors, first RE buyouts, gov integrations, audits

Section 08

Risk Management

Risk	Severity	Mitigation
Smart contract exploit	High	CertiK + Hacken audits; OpenZeppelin patterns; bug bounty
GCSC price volatility	Medium	GCST stablecoin for disbursements; dynamic burn mechanics
Regulatory action (lending)	Medium	Licensed partner compliance layer; jurisdiction-specific rollout
AI agent error	Medium	Loans >0K require DAO human approval; RAA confidence threshold
Low liquidity at launch	Medium	10% bootstrap allocation; Proton Loan backstop

Disclaimer: This whitepaper is for informational purposes only and does not constitute financial, legal, or investment advice. GCSC tokens involve risk. Consult qualified advisors before participating.